

CHAPTER C31

WIND TUNNEL PROCEDURE

Wind tunnel testing is specified when a building or other structure contains any of the characteristics defined in Sections 27.1.3, 28.1.3, 29.1.3, or 30.1.3 or when the designer wishes to more accurately determine the wind loads. For some building or structure shapes, wind tunnel testing can reduce the conservatism caused by enveloping of wind loads inherent in the Directional Procedure, Envelope Procedure, or Analytical Procedure for Components and Cladding (C&C). Also, wind tunnel testing accounts for shielding or channeling and can more accurately determine wind loads for a complex building or structure shape than can the Directional Procedure, Envelope Procedure, or Analytical Procedure for C&C. It is the intent of the standard that any building or other structure can be allowed to use the wind tunnel testing method to determine wind loads. Requirements for proper testing are given in ASCE 49 (2012).

It is common practice to resort to wind tunnel tests when design data are required for the following wind-induced loads:

1. Curtain wall pressures resulting from irregular geometry;
2. Across-wind and/or torsional loads;
3. Periodic loads caused by vortex shedding; and
4. Loads resulting from instabilities, such as flutter or galloping.

Boundary-layer wind tunnels capable of developing flows that meet the conditions stipulated in Section 31.2 typically have test-section dimensions in the following ranges: width of 6 to 12 ft (2 to 4 m), height of 6 to 10 ft (2 to 3 m), and length of 50 to 100 ft (15 to 30 m). Maximum wind speeds are ordinarily in the range of 25 to 100 mi/h (10 to 45 m/s). The wind tunnel may be either an open-circuit or closed-circuit type.

Three basic types of wind tunnel test models are commonly used. These are designated as follows: (1) rigid pressure model (PM), (2) rigid high-frequency base balance model (H-FBBM), and (3) aeroelastic model (AM). One or more of the models may be used to obtain design loads for a particular building or structure. The PM provides local peak pressures for design of elements, such as cladding and mean pressures, for the determination of overall mean loads. The H-FBBM measures overall fluctuating loads (aerodynamic admittance) for the determination of dynamic responses. When motion of a building or structure influences the wind loading, the AM is used for direct measurement of overall loads, deflections, and accelerations. Each of these models, together with a model of the surroundings (proximity model), can provide information other than wind loads, such as snow loads on complex roofs, wind data to evaluate environmental impact on pedestrians, and concentrations of air pollutant emissions for environmental impact determinations. Several references provide detailed information and guidance for the determination of wind loads and other types of design data

by wind tunnel tests (Cermak 1977; Reinhold 1982; ASCE 1999; Boggs and Peterka 1989).

Wind tunnel tests frequently measure wind loads that are significantly lower than required by Chapters 26, 27, 28, 29, and 30 because of the shape of the building or other structure; the likelihood that the highest wind speeds occur at directions where the building or structure's shape or pressure coefficients are less than their maximum values; specific buildings or structures included in a detailed proximity model that may provide shielding in excess of that implied by exposure categories; and necessary conservatism in enveloping load coefficients in Chapters 28 and 30. In some cases, adjacent buildings or structures may shield the subject building or structure sufficiently that removal of one or two of the adjacent buildings or structures could significantly increase wind loads. Additional wind tunnel testing without specific nearby buildings or structures (or with additional buildings or structures if they might cause increased loads through channeling or buffeting) is an effective method for determining the influence of adjacent buildings or structures.

For this reason, the standard limits the reduction that can be accepted from wind tunnel tests to 80% of the result obtained from Part 1 of Chapter 27 or Part 1 of Chapter 28, or Chapter 30, if the wind tunnel proximity model included any specific influential buildings or other objects that, in the judgment of an experienced wind engineer, are likely to have substantially influenced the results beyond those characteristic of the general surroundings. If there are any such buildings or objects, supplemental testing can be performed to quantify their effect on the original results and possibly justify a limit lower than 80%, by removing them from the detailed proximity model and replacing them with characteristic ground roughness consistent with the adjacent roughness. A specific influential building or object is one within the detailed proximity model that protrudes well above its surroundings, or is unusually close to the subject building, or may otherwise cause substantial sheltering effect or magnification of the wind loads. When these supplemental test results are included with the original results, the acceptable results are then considered to be the higher of both conditions.

However, the absolute minimum reduction permitted is 65% of the baseline result for C&C and 50% for the main wind force resisting system (MWFRS). A higher reduction is permitted for MWFRS because C&C loads are more subject to changes caused by local channeling effects when surroundings change, and they can easily be dramatically increased when a new adjacent building is constructed. It is also recognized that cladding failures are much more common than failures of the MWFRS. In addition, for the case of MWFRS, it is easily demonstrated that the overall drag coefficient for certain common building shapes, such as circular cylinders (especially with rounded or domed

tops), is one-half or less of the drag coefficient for the rectangular prisms that form the basis of Chapters 27, 28, and 30.

For C&C, the 80% limit is defined by the interior Zones 1 and 4 in Figs. 30.3-1, 30.3-2A–C, 30.3-3, 30.3-4, 30.3-5A–B, 30.3-6, 30.3-7, and 30.4-1. This limitation recognizes that pressures in the edge zones are the ones most likely to be reduced by the specific geometry of real buildings compared with the rectangular prismatic buildings assumed in Chapter 30. Therefore, pressures in edge and corner zones are permitted to be as low as 80% of the interior pressures from Chapter 30 without the supplemental tests. The 80% limit based on Zone 1 is directly applicable to all roof areas, and the 80% limit based on Zone 4 is directly applicable to all wall areas.

The limitation on MWFRS loads is more complex because the load effects (e.g., member stresses or forces, deflections) at any point are the combined effect of a vector of applied loads instead of a simple scalar value. In general, the ratio of forces or moments or torques (force eccentricity) at various floors throughout the building using a wind tunnel study will not be the same as those ratios determined from Chapter 27 and 28, and therefore comparison between the two methods is not well defined. Requiring each load effect from a wind tunnel test to be no less than 80% of the same effect resulting from Chapters 27 and 28 is impractical and unnecessarily complex and detailed, given the approximate nature of the 80% value. Instead, the intent of the limitation is effectively implemented by applying it only to a simple index that characterizes the overall loading. For flexible (tall) buildings, the most descriptive index of overall loading is the base overturning moment. For other buildings, the overturning moment can be a poor characterization of the overall loading, and the base shear is recommended instead.

C31.4 LOAD EFFECTS

C31.4.1 Mean Recurrence Intervals of Load Effects. Examples of analysis methods for combining directional wind tunnel data with the directional meteorological data or probabilistic models based thereon are described in Lepage and Irwin (1985), Rigato et al. (2001), Isyumov et al. (2013), Irwin et al. (2005), Simiu and Filliben (2005), and Simiu and Miyata (2006).

C31.4.2 Limitations on Wind Speeds. Section 31.4.2 specifies that the statistical methods used to analyze historical wind speed and direction data for wind tunnel studies shall be subject to the same limitations specified in Section 31.4.2 that apply to the Analytical Method.

Database-Assisted Design. Wind tunnel aerodynamics databases that contain records of pressures measured synchronously at large numbers of locations on the exterior surface of building models have been developed by wind researchers, such as Simiu et al. (2003) and Main and Fritz (2006). Such databases include data that permit a designer to determine, without specific wind tunnel tests, wind-induced forces and moments in MWFRSs and C&C of selected shapes and sizes of buildings. A public domain set of such databases, recorded in tests conducted at the University of Western Ontario (Ho et al. 2005; St. Pierre et al. 2005) for buildings with gable roofs is available on the National Institute of Standards and Technology website, www.nist.gov/wind (NIST 2012). Interpolation software for buildings with similar shape and with dimensions close to and intermediate between those included in the set of databases is also available on that site. Because the database results are for generic surroundings as permitted in ASCE 49 interpolation or extrapolation from these databases should be used only if Condition 2 of Section 27.1.2 is

true. Extrapolations from available building shapes and sizes are not permitted, and interpolations in some instances may not be advisable. For these reasons, the guidance of an engineer experienced in wind loads on buildings and familiar with the usage of these databases is recommended.

All databases must have been obtained using testing methodology that meets the requirements for wind tunnel testing specified in Chapter 31.

C31.4.3 Wind Directionality. The variability of wind speed determined for particular azimuth intervals is greater than that of the wind speed determined regardless of wind direction (Isyumov et al. 2013). Consequently, wind loads and wind-induced effects determined by allowing for wind directionality are inherently less certain. Several methods for combining data from wind tunnel model studies with information on wind speed and direction at the project site are currently in use (Isyumov et al. 2013; Yeo and Simiu 2011; and Simiu 2011). Whichever method is used shall be clearly described to allow scrutiny by the designer and the Authority Having Jurisdiction. A common approach for allowing for uncertainties in the wind direction is to rotate the project wind climate relative to the orientation of the building or structure. This rotation of the wind climate at the building location is intended to ensure that the wind loads determined for design are not unconservative and shall be considered regardless of the method used for arriving at the design wind speeds. The appropriate magnitude of wind climate rotation varies depending on the quality and resolution of the directional wind climate data at the project site.

C31.6 ROOF-MOUNTED SOLAR COLLECTORS FOR ROOF SLOPES LESS THAN 7 DEGREES

C31.6.1 Wind Tunnel Test Requirements. For solar collector installations, it is necessary to model a generic building with the solar collectors on the roof of a scaled building, then generate (GC_m) pressure coefficients that are applicable to any site, a wide range of building sizes, and varied collector layouts. The approach needs to be similar to that used to develop the (GC_p) figures in ASCE 7 by modeling the generic buildings with various features to capture a wide range of effects. The objective of such testing is to evaluate aerodynamic effects accounted for by (GC_m) pressure coefficients (in contrast to site-specific wind tunnel testing, which also evaluates the effect of surrounding structures and terrain). Nearby buildings should not be included unless they are to be a part of every design application for this collector.

Wind tunnel testing for roof-mounted solar collectors must include a sufficiently large test matrix to address an appropriate range of the relevant variables that affect wind loads as listed in the provisions. Tests are often performed at model scale of 1:50 or larger where the match of wind tunnel turbulence characteristics is not ideal, resulting in some added requirements for testing, including integral scale limits. The wind tunnel study should provide recommendations for setback distances from larger rooftop equipment, penthouses, clerestories, and other building features. Guidance for testing is provided in ASCE 49, Kopp and Banks (2013), and Kopp et al. (2011, 2012). Wind loads are expressed as coefficients usable in Chapters 27, 29, and 30 to produce loads in engineering units. Alternately, a different formulation of nondimensional load coefficients may be used provided that the analysis procedure is clearly defined in the test report.

C31.6.1.1 Limitations on Wind Loads for Rooftop Solar Collectors. The minimum components and cladding wind

load pressures indicated in ASCE 7 are primarily applicable to the building envelope and are not entirely applicable to rooftop solar collectors. The limitations contained herein are to establish the lower bound wind pressures for wind tunnel studies of conditions similar to those addressed by Fig. 29.4-7. The limits on wind tunnel results shown in Fig. 29.4-7 represent an envelope of wind loads measured in the wind tunnel without deflectors or shrouds that are commonly used to lower wind loads. Specific installations or collector geometries may give significantly lower loads than Fig. 29.4-7; limits are imposed to prevent too much deviation from the enveloped results.

C31.6.1.2 Peer Review Requirements for Wind Tunnel Tests of Roof-Mounted Solar Collectors. Solar collector systems that have aerodynamic devices or more efficient profiles can have wind tunnel based wind loads less than the lower bound thresholds indicated in Sections 31.6.1 and 31.6.2. In order to use these lower values, a peer review of the test and report is required. The peer reviewer qualifications and requirements are included to promote consistencies among the various jurisdictions so that a peer review could be accepted by multiple enforcement agencies. The peer review qualifications are intended to be those of a wind tunnel expert familiar with wind tunnel testing of buildings and the applicability of the ASCE 7 provisions to determine generalized wind design coefficients for roof-mounted solar collectors. One source for peer reviewers is the American Association for Wind Engineering's (AAWE) boundary layer wind tunnels list (http://www.aawe.org/info/wind_tunnels.php).

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